ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** an acid	Acid.
2	**(b)** 40 mL	Proportional.
3	**(a)** acidic	Acid rain.
4	**(d)** an acid	An acid neutralises the base.
5	**(b)** 30 mL	Doubling.
6	**(d)** it is a base used to relieve acidity	Carbonates react with acids.
7	**(a)** turns almost neutral	They neutralise.
8	**(b)** 500 kg	200 x 2.5.
9	**(a)** a carbonate and an acid	CO2 is released.
10	**(c)** red, purple, blue	Acid to base.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
12	Use litmus or turmeric (1); Different colours appear (1).	1+1
13	Neither acid nor base (1); The effects of acid and base cancel out (1).	1+1
14	A mucus layer protects it (1); The acid is dilute (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
16	Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
17	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1
	OR: Add slaked lime (1); Add wood ash (1); Add powdered limestone (1).	1+1+1
18	Neutralisation releases heat (1); It is exothermic (1); Salt and water form (1).	1+1+1
	OR: Add equal juice (1); Compare the red colour depth (1); More intense means more acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dissolve equal tablets (1); Add to same volume of acid with indicator (1); Count amount needed to neutralise (1); Same conditions (1); Most effective needs the least acid change (1).	1+1+1+1+1
	OR: Some crops prefer acidic soil (1); Others basic (1); Lime reduces acidity (1); Manure improves soil (1); Test regularly (1).	1+1+1+1+1
20	Soil treatment (1); Antacids (1); Acid rain (1); Treating wastes (1); Principle: acid + base gives salt + water (1).	1+1+1+1+1
	OR: Natural indicators are safe (1); Cheap (1); Less precise (1); Lab indicators precise (1); Choice by need (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the acid was stronger or more concentrated	Excess acid.
2	**(b)** acid-carbonate reaction	CO2.
3	**(d)** $HCl + NaOH \longrightarrow NaCl + H_2O$	Neutralisation.
4	**(d)** acidic	Red is acid.
5	**(d)** vinegar	Acetic acid.
6	**(c)** aqueous $NaOH$	Sodium hydroxide.
7	**(c)** sugar solution	Neutral.
8	**(c)** a paste of baking soda and water	A base neutralises the acid.
9	**(c)** the acid dissolves copper compounds	Citric acid.
10	**(d)** magnesium hydroxide	Base.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is basic (1); Bases feel soapy (1).	1+1
12	Sodium hydrogencarbonate (baking soda) (1); Mild base (1).	1+1
13	To avoid accidents and spills (1); So that nobody uses it by mistake (1).	1+1
14	It makes the water acidic (1); Fish and plants cannot survive in it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Baking soda is a base (1); Reacts with acid (1); Releases carbon dioxide (1).	1+1+1
16	Add lime or another base (1); Test until neutral (1); Then release (1).	1+1+1
17	Baking soda (1); It is a base (1); Others acidic (1).	1+1+1
	OR: Bacteria make acid from sugar (1); Acid dissolves enamel (1); Brush with basic paste (1).	1+1+1
18	Neutralises formic acid (1); Use litmus (1); No colour change (1).	1+1+1
	OR: Soap turns it red (1); Lemon turns it yellow again (1); Acid neutralises (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prepare indicator solutions (1); Test each substance (1); Record colours (1); Limitations: colour not exact (1); Safe handling (1).	1+1+1+1+1
	OR: Contains baking soda and an acid (1); Mixed with water (1); Carbon dioxide formed (1); Dough rises (1); Neutral products (1).	1+1+1+1+1
20	Collect rain in clean glass (1); Test with litmus (1); Compare with distilled water (1); Repeat in several places (1); Conclude and report (1).	1+1+1+1+1
	OR: Use litmus (1); Compare with tap water (1); If acidic, inform adults (1); Do not drink (1); Treat water (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation	Salt remains.
2	**(d)** washing soda is basic	Base turns it green.
3	**(b)** making an acid stronger	Opposite use.
4	**(d)** dilute hydrochloric acid	Acid.
5	**(d)** acid and baking soda	Carbonates with acids.
6	**(c)** carbon dioxide	CO2.
7	**(b)** acid neutralises the base	Neutralisation.
8	**(a)** it is corrosive	Burns tissues.
9	**(a)** turns lime water milky	Standard test.
10	**(d)** the mixture can become warm because heat is released	Neutralisation is exothermic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bases dissolve grease (1); Soapy (1).	1+1
12	$H_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4 + 2\,H_2O$ (1); Calcium sulphate and water (1).	1+1
13	Lime is basic (1); Reduces acidity (1).	1+1
14	Bacteria multiply faster (1); More lactic acid forms (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Deposit is carbonate (1); Acid reacts with it (1); Gas is released (1).	1+1+1
16	It does not change litmus (1); It is neither acid nor base (1); Pure water is neutral (1).	1+1+1
17	Acidic (1); Damage teeth (1); Add sugar (1).	1+1+1
	OR: Litmus unchanged (1); Turmeric unchanged (1); Compare with water (1).	1+1+1
18	Strong bases are corrosive (1); They can burn the skin (1); Gloves protect the hands (1).	1+1+1
	OR: Test with red cabbage (1); More green means more basic (1); Compare (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add litmus to acid: red (1); Add base slowly (1); At neutral violet (1); $HCl + NaOH \longrightarrow NaCl + H_2O$ (1); Goggles and gloves (1).	1+1+1+1+1
	OR: Burns (1); Rinse with plenty of water (1); Remove contaminated clothing (1); Medical help (1); Do not use strong neutralisers (1).	1+1+1+1+1
20	Chemical bonds change (1); Heat released (1); Controlled by slow addition (1); Strong acid and base must be mixed with care (1); Danger if strong acid and base mix (1).	1+1+1+1+1
	OR: Safe for skin (1); Effective cleaning (1); Too strong harms (1); Balanced formulas (1); Testing by manufacturers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.